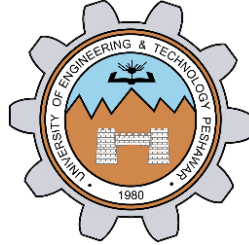


LAB # 05

Interfacing Seven Segment Display to 8051 Development kit



Spring 2024

CSE-307L Microprocessor Based System Design Lab

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Class Section: **C**

“On my honor, as student at University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

Submitted to:

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Objectives of The Lab:

- Understand the concept of 7-segment display.
- interfacing 7-segment display with 8051 microcontroller

7-segment Display:

The 7-segment display, also written as “seven segment display”, consists of seven LEDs (hence its name) arranged in a rectangular fashion as shown. Each of the seven LEDs is called a segment because when illuminated the segment forms part of a numerical digit to be displayed.



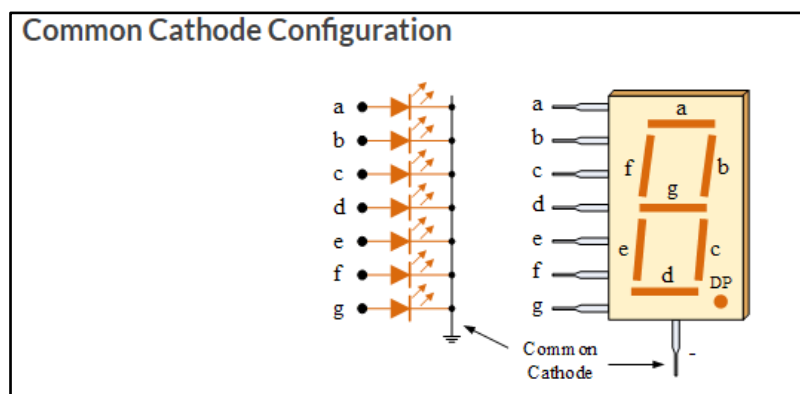
Two Types of LED 7-Segment:

two types of LED 7-segment display called:

Common Cathode (CC) and Common Anode (CA).

Common Cathode:

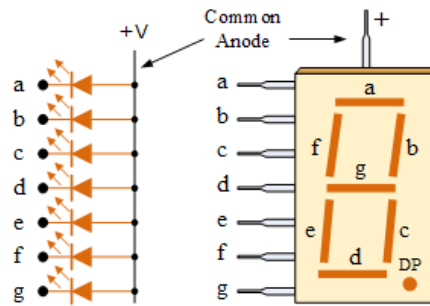
The Common Cathode (CC) – In the common cathode display, all the cathode connections of the LED segments are joined together to logic “0” or ground. The individual segments are illuminated by application of a “HIGH”, or logic “1” signal via a current limiting resistor to forward bias the individual Anode terminals (a-g).



Common Anode:

The Common Anode (CA) – In the common anode display, all the anode connections of the LED segments are joined together to logic “1”. The individual segments are illuminated by applying a ground, logic “0” or “LOW” signal via a suitable current limiting resistor to the Cathode of the particular segment (a-g).

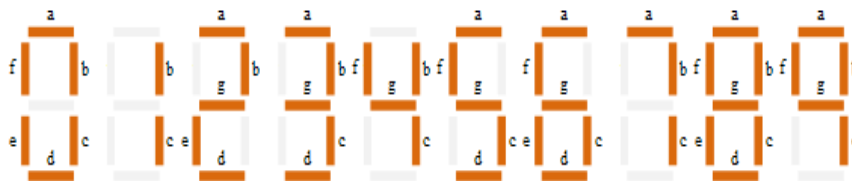
Common Anode Configuration



Displaying Numbers:

Each segment has label and can be turned off and on. turning on different combination of led segments shows different numbers.

Digital Segments for all Numbers



Lab Total Tasks:

- 1) Run all the program given in the lecture
 - 2) Display different digits on the Seven Segment Display
 - 3) Display digits from 0 to F on Seven Segment Display
- Hint: Use 7-segment common Anode

Task 1:

Run all the program given in the lecture.

C Code:

```

1 #include<reg51.h>
2 void delay();
3
4 void main(void) {
5
6     while(1){
7         P0 = 0x00;
8
9         for(;;){
10            P0 = 0xF9;
11            delay();
12        }
13    }
14 }
15 void delay(){
16     unsigned int y;
17     for(y = 0; y<30000; y++){
18     }
19 }
20

```

Keil Simulation:

The Keil uVision simulation interface displays the following components:

- Registers Window:**

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x0d
a	0x00
b	0x00
sp	0x09
sp_max	0x09
dptr	0x000
PC	0x001
states	463
sec	0.000
psw	0x80
- Disassembly Window:**

```

18:      }
C:0x0803 0F      INC      R7
C:0x0804 BF0001  CJNE     R7,#0x00,C:0808
C:0x0807 0F      TNC

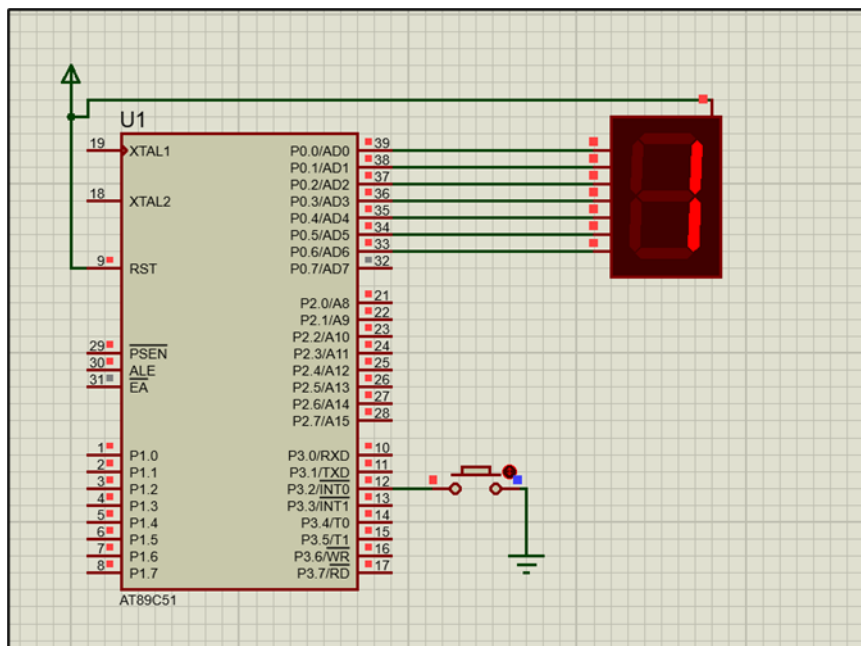
```
- Source Code Window (task1.c):**

```

6  while(1){
7      P0 = 0x00;
8
9      for(;;){
10         P0 = 0xF9;
11         delay();
12     }
13 }
14
15 void delay(){
16     unsigned int y;
17     for(y = 0; y<30000; y++){
18     }
19 }
20

```

Proteus Simulation:



Task 2:

Display different digits on the Seven Segment Display

C Code:

```
1  #include<reg51.h>
2  void delay();
3
4  void main(void) {
5
6      while(1){
7          P0 = 0x40; //0
8          delay();
9          P0 = 0x79; //1
10         delay();
11         P0 = 0x24; //2
12         delay();
13         P0 = 0x30; //3
14         delay();
15         P0 = 0x19; //4
16         delay();
17         P0 = 0x12; //5
18         delay();
19         P0 = 0x02; //6
20         delay();
21         P0 = 0x78; //7
22         delay();
23         P0 = 0x00; //8
24         delay();
25         P0 = 0x18; //9
26     }
27 }
28 void delay(){
29     unsigned int y, x;
30     for(x = 0; x<30000; x++);
31     for(y = 0; y<30000; y++);
32 }
33
```

Keil Simulation:

Disassembly

```

9:          P0 = 0x79; //1
C:0x0806   758079  MOV     P0(0x80),#0x79
10:         delay();
C:0x0809   12083B  LCALL   delay(C:083B)

```

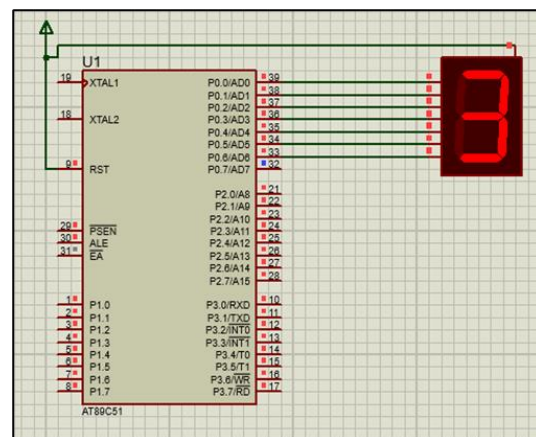
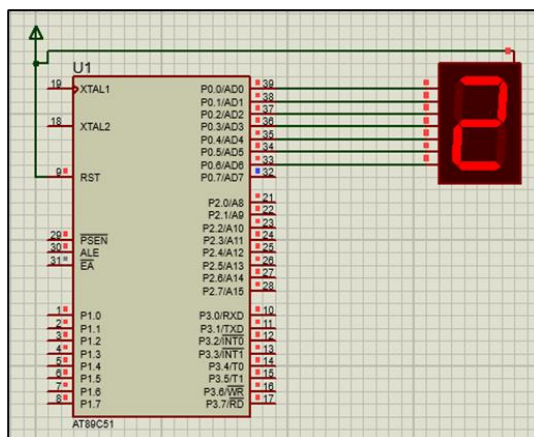
task3.c

```

4 void main(void) {
5
6     while(1){
7         P0 = 0x40; //0
8         delay();
9         P0 = 0x79; //1
10        delay();
11        P0 = 0x24; //2
12        delay();
13        P0 = 0x30; //3
14        delay();
15        P0 = 0x19; //4
16        delay();
17        P0 = 0x12; //5
18        delay();
19        P0 = 0x02; //6

```

Proteus Simulation:



Task 3:

Display digits from 0 to F on Seven Segment Display

C Code:

```
1  #include<reg51.h>
2  void delay();
3
4  void main(void) {
5
6      while(1){
7          P0 = 0x40; //0
8          delay();
9          P0 = 0x79; //1
10         delay();
11         P0 = 0x24; //2
12         delay();
13         P0 = 0x30; //3
14         delay();
15         P0 = 0x19; //4
16         delay();
17         P0 = 0x12; //5
18         delay();
19         P0 = 0x02; //6
20         delay();
21         P0 = 0x78; //7
22         delay();
23         P0 = 0x00; //8
24         delay();
```

```
24         delay();
25         P0 = 0x18; //9
26         delay();
27         P0 = 0x08; //A
28         delay();
29         P0 = 0x00; //B
30         delay();
31         P0 = 0x46; //C
32         delay();
33         P0 = 0x40; //D
34         delay();
35         P0 = 0x06; //E
36         delay();
37         P0 = 0x0E; //F
38         delay();
39
40     }
41 }
42 void delay(){
43     unsigned int y, x;
44     for(x = 0; x<30000; x++);
45     for(y = 0; y<30000; y++);
46 }
47
```

Keil Simulation:

Disassembly

```

12:          delay();
C:0x080C  758024  MOV     P0 (0x80), #0x24
13:          P0 = 0x30; //3
C:0x080F  120862  LCALL   delay(C:0862)

```

task2.c

```

8  delay();
9  P0 = 0x79; //1
10 delay();
11 P0 = 0x24; //2
12 delay();
13 P0 = 0x30; //3
14 delay();
15 P0 = 0x19; //4
16 delay();
17 P0 = 0x12; //5
18 delay();
19 P0 = 0x02; //6
20 delay();
21 P0 = 0x78; //7

```

Parallel Port 0

Port 0: 0x79 7 Bits 0

Pins: 0x40

Proteus Simulation:

